SCOOT · Level 2 · Advanced

Three small optimisers that run a whole city's lights

How SCOOT turns its link models into coordinated splits, offsets and cycle times, and copes with jams, buses and broken detectors.

smrahmanbd.github.io/scoot-learning-lab

8 ideas, each building on the last

1 Regions, nodes and double cycling

40 → 17 s pedestrian wait when a 96 s crossing double cycles

2 The split optimiser

88% split decisions that were 'keep' in our SUMO run

3 The offset optimiser

230 → 94 PI after four offset steps in the worked example

4 The cycle optimiser

40 → 88 s cycle growth in the worked peak

5 Congestion, exit blocking and gating

0.85 → 1.02 weighted DoS with 20% congestion, importance 1

6 Bus priority: extensions and recalls

22.9 s mean bus delay with priority (34.7 s without)

7 The wider SCOOT system

2020 SCOOT 7: cooperative data and cloud-first design

8 The full system on a SUMO region

−65% delay vs fixed time on the SUMO region

Three surprises to start

Most decisions change nothing.

In our SUMO run the split optimiser made 846 decisions and chose 'keep the timing' 747 times, about 88%. TRL's original description says about two thirds. Doing nothing, most of the time, is a sign the plan is already about right.

The green wave is never designed.

Nobody tells SCOOT the best offset between two junctions. It tries 4 s earlier and 4 s later once per cycle and keeps whichever its model likes best. A few minutes later the platoons arrive on green.

Switching one thing off can help.

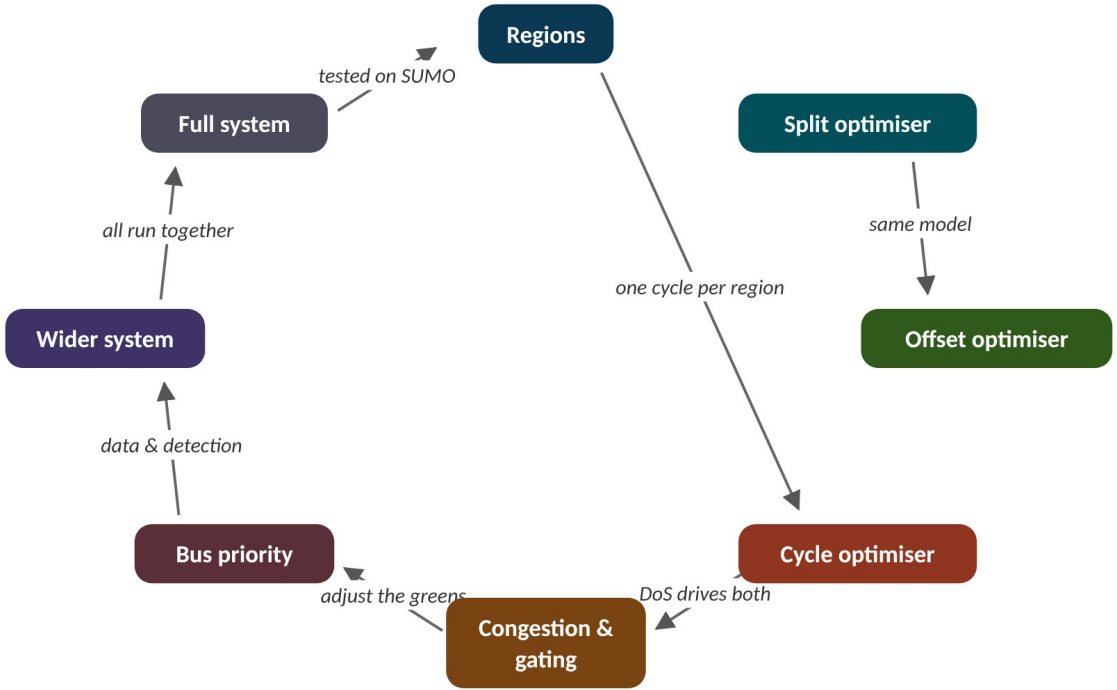
In our ablation study, freezing the cycle at 60 s gave slightly lower delay (26.8 s) than full SCOOT (30.4 s). The reason teaches you why engineers set a maximum cycle for every region.

Words you will meet

Region	A group of junctions that all use the same cycle time, so they can be coordinated.
Node	One junction or crossing inside a region.
Stage change	The moment one stage's green ends and the next stage's starts.
DoS	Degree of saturation: how full a link's green is. SCOOT aims for 90% or less.
PI	Performance index: delay + K × stops. Lower is better.

Optimiser	A small rule that tries a few options in the model and keeps the best one.
Temporary / permanent change	A change applied to this cycle only, or stored in the plan for the future.
Critical node	The busiest junction in a region; it decides the region's cycle time.
Gating	Holding traffic back on purpose, where queues do no harm, to protect a bottleneck.
AVL / SVD	Ways to detect buses: automatic vehicle location (GPS) or a selective vehicle detector.

The map of today's ideas



1 Regions, nodes and double cycling

IN PLAIN WORDS

An orchestra plays to one tempo so everyone stays together. A triangle player who has little to do can play two notes per beat and still stay in time. That is double cycling: a quiet node runs twice as fast but stays in step with its region.

- A region shares one cycle time
- Shared cycle makes offsets meaningful
- Quiet nodes may double cycle
- Only if DoS stays below 90%

40 → 17 s

pedestrian wait when a 96 s crossing double cycles

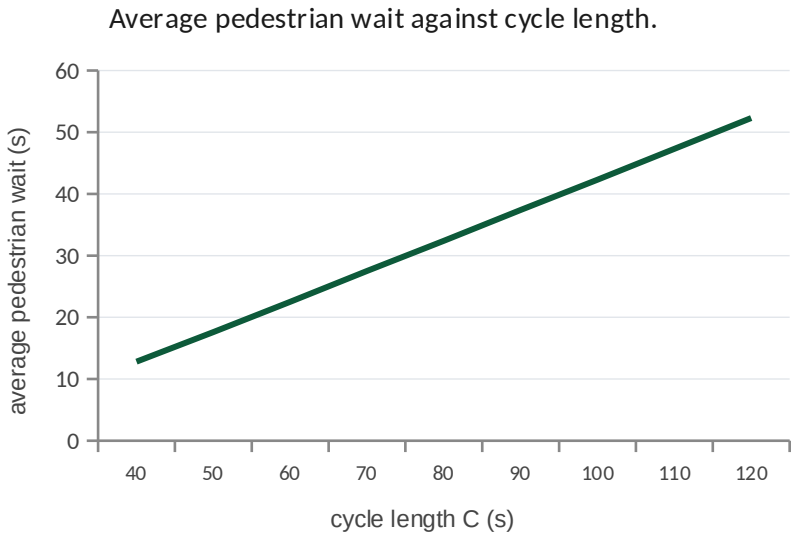
Worked example: Should this crossing double cycle?

$$\text{average pedestrian wait} \approx (C - g)^2 \div 2C$$
$$\text{double cycle allowed if DoS at } C/2 \le 0.90$$

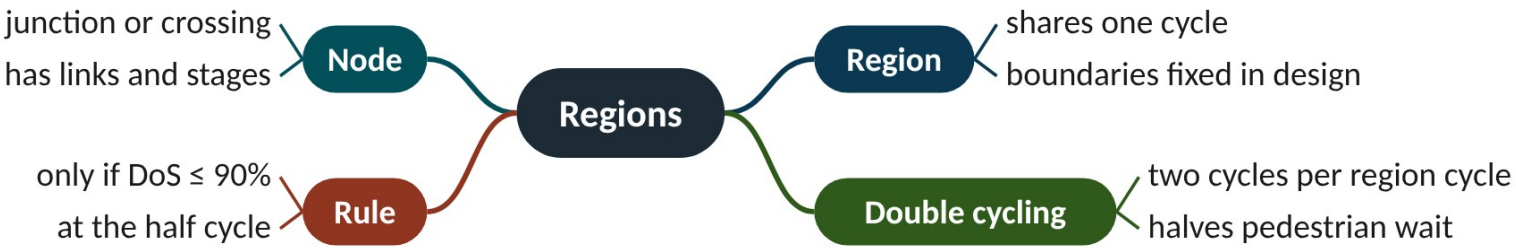
Region cycle C = 96 s. Pedestrian green g = 8 s. Vehicle demand 0.3 LPU/s, SATO 1.2 LPU/s. At C/2 = 48 s the vehicles get 30 s of effective green.

1. Wait at 96 s: $(96 - 8)^2 \div (2 \times 96) = 7744 \div 192 =$ **40.3 s**.
2. Wait at 48 s: $(48 - 8)^2 \div (2 \times 48) = 1600 \div 96 =$ **16.7 s**.
3. Vehicle demand per 48 s cycle: $0.3 \times 48 =$ **14.4 LPU**.
4. Capacity: $30 \times 1.2 =$ **36 LPU**. DoS $= 14.4 \div 36 =$ **0.40**.
5. $0.40 \le 0.90$, so double cycling is **allowed**.

Double cycling cuts the average pedestrian wait from about 40 s to about 17 s while vehicles still have plenty of spare green.



Remember it: Regions



PICTURE THIS · STOP 1: STAGE DOOR

At the stage door the whole orchestra marches in, every foot landing on one giant metronome's tick. A tiny triangle player at the back skips two steps for every one tick, grinning, never falling out of time.

TRAP

Double cycling does not give the crossing its own independent cycle. It runs exactly two cycles per region cycle, so it stays coordinated. A node with a truly different cycle would belong to another region.

2 The split optimiser

IN PLAIN WORDS

Two children share a seesaw that should stay level. Every few seconds a helper may move a 4 kg bag to one side, or leave it. To remember the lesson, the helper also glues 1 kg to that side for good.

- Before every stage change: −4, 0 or +4
- Score each option by its worst DoS
- Temporary ±4 s, permanent ±1 s
- Most decisions are 'keep'

88%

split decisions that were 'keep' in our SUMO run

Worked example: Choose −4, 0 or +4

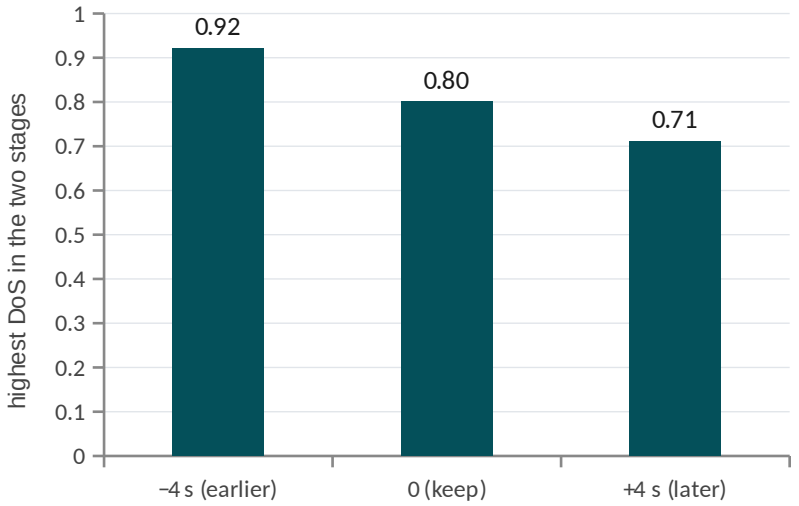
score(option) = highest DoS among links in both stages
DoS = demand per cycle ÷ (green × SATO)

Two stages, 60 s cycle, 50 s of green in total. Now: stage A 30 s, stage B 20 s. Demand A 24 LPU, B 10 LPU per cycle. SATO 1.0 LPU/s for both.

1. −4 (A 26, B 24): DoS A = 24 ÷ 26 = 0.92; B = 10 ÷ 24 = 0.42; highest **0.92**.
2. 0 (A 30, B 20): A = 24 ÷ 30 = 0.80; B = 10 ÷ 20 = 0.50; highest **0.80**.
3. +4 (A 34, B 16): A = 24 ÷ 34 = 0.71; B = 10 ÷ 16 = 0.63; highest **0.71**.
4. Lowest highest DoS: **+4**. A runs 34 s this cycle.
5. Permanent change: stored green A goes from 30 to **31 s**.

Stage A is the busier one, so SCOOT gives it 4 s more now and 1 s more for the future. If demand stays, the stored plan keeps creeping towards balance.

Highest DoS for each option in the worked example.



Remember it: Split optimiser



PICTURE THIS · STOP 2: SEESAW

In the green room two singers sit on a seesaw. A stagehand stands ready with a 4 kg sandbag; every few seconds he drops it on one end, or shrugs and keeps it. Each time he also glues a single 1 kg weight to that end for ever.

TRAP

'+4' does not mean 'add 4 s to every stage'. It moves one stage change 4 s later: the ending stage gains 4 s and the starting stage loses 4 s, so the cycle length stays the same.

3 The offset optimiser

IN PLAIN WORDS

Imagine finding the lowest point of a valley in thick fog. You can only feel the ground one step to the left and one step to the right. You step towards whichever is lower, again and again, until both sides are higher. You are at the bottom.

- Once per cycle: node −4, 0 or +4
- Score: PI on links in and out
- Walks downhill to the best offset
- Finds the green wave by itself

230 → 94

PI after four offset steps in the worked example

Worked example: Walking down the PI curve

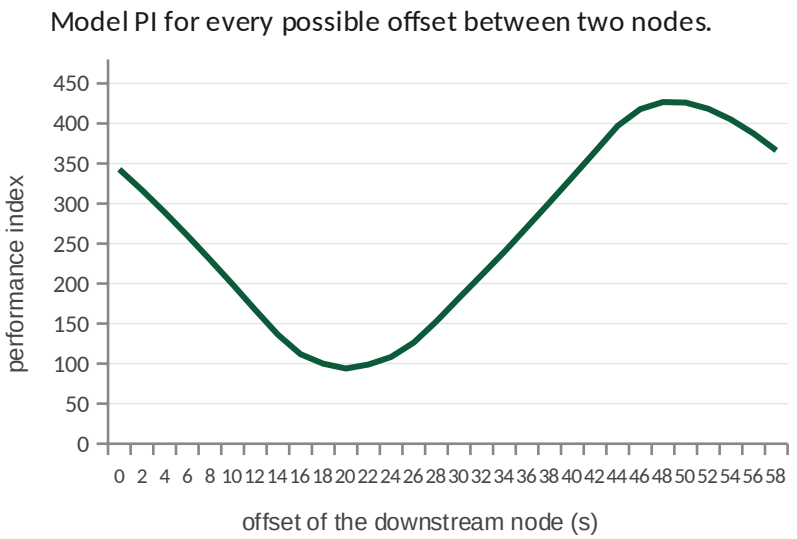
$$PI(\text{offset}) = \sum_{\text{links in and out}} (\text{delay} + K \times \text{stops})$$

$$\text{new offset} = \text{best of } \{ \text{offset} - 4, \text{offset}, \text{offset} + 4 \}$$

Two nodes 300 m apart (cruise time 22 s), 60 s cycle. The model's PI at each offset is: 4 s → 289, 8 s → 230, 12 s → 167, 16 s → 112, 20 s → 94, 24 s → 108. The node starts at 8 s.

1. Cycle 1, at 8 s: options 4/8/12 give 289/230/167. Lowest: **12 s**.
2. Cycle 2, at 12 s: 230/167/112. Move to **16 s**.
3. Cycle 3, at 16 s: 167/112/94. Move to **20 s**.
4. Cycle 4, at 20 s: 112/94/108. The middle is lowest: **keep 20 s**.

In four cycles (four minutes) the offset found the bottom of the curve, and PI fell from 230 to 94. SCOOT never needed to know the distance or the speed.



Remember it: Offset optimiser



PICTURE THIS · STOP 3: ECHO CORRIDOR

In a long corridor two drummers try to make their beats meet in the middle. Each cycle the second drummer shuffles 4 steps forward or back, watching a glowing noise meter on the wall, and stops when the meter is at its lowest.

TRAP

The offset optimiser uses PI (delay and stops), not DoS. DoS hardly changes with offset, because offset moves when the green happens, not how long it is.

4 The cycle optimiser

IN PLAIN WORDS

A conductor sets the tempo for the whole orchestra. If the busiest section cannot fit its notes in, the conductor slows down a little. If everyone has time to spare, the conductor speeds up a little. The tempo changes only every few minutes, so the players are never confused.

- Every 5 minutes, 2.5 while rising
- Critical node decides for the region
- Above 90% longer, below 80% shorter
- Steps of 4, 8 or 16 s

40 → 88 s

cycle growth in the worked peak

Worked example: A rising peak

$$\text{critical DoS} \approx Y \times C \div (C - L)$$
$$\text{step} = 4\text{ s } (C < 64),\ 8\text{ s } (C < 128),\ 16\text{ s } (C \geq 128)$$

Critical node: lost time L = 12 s, flow ratio Y = 0.78 in the peak. The cycle is 40 s.

1. DoS = 0.78 × 40 ÷ (40 − 12) = 31.2 ÷ 28 = **1.11**. Above 0.90 → up 4 s to **44 s**.

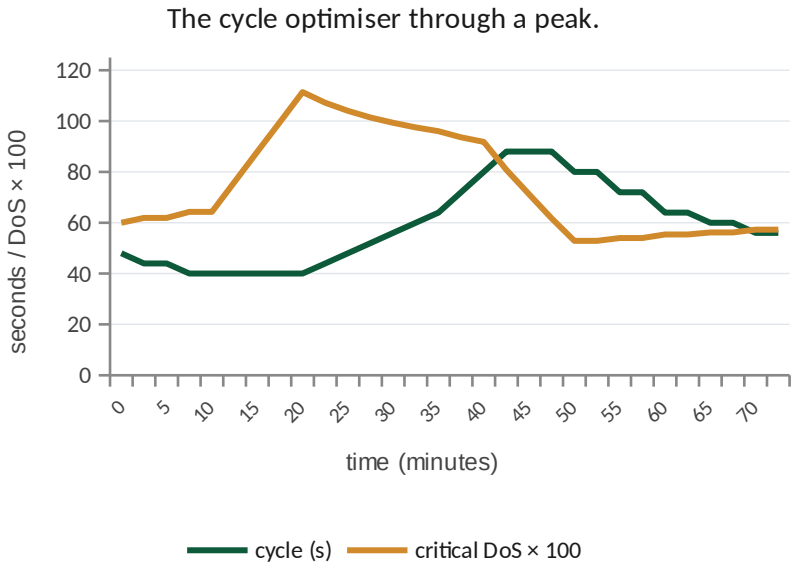
2. 2.5 min later: 0.78 × 44 ÷ 32 = **1.07**. Still above → **48 s**.

3. At 64 s: 0.78 × 64 ÷ 52 = **0.96**. The step is now 8 s → **72 s**.

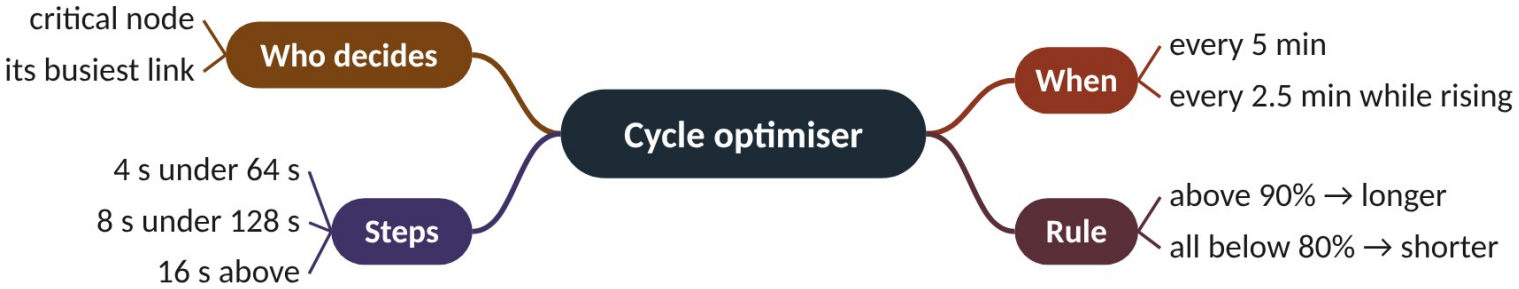
4. At 80 s: 0.78 × 80 ÷ 68 = **0.92** → **88 s**.

5. At 88 s the peak eases (Y = 0.70): 0.70 × 88 ÷ 76 = **0.81**. Between 0.80 and 0.90 → **hold**.

Going up takes small steps every 2.5 minutes. Coming down happens only every 5 minutes, so the region can spend a while on a cycle longer than it needs.



Remember it: Cycle optimiser



PICTURE THIS · STOP 4: METRONOME

On the podium stands a metronome as tall as a giraffe. Three weights hang on it, stamped 4, 8 and 16. Every five minutes a cuckoo pops out; the conductor glances at a gauge painted red above 90 and slides one weight up or down.

TRAP

A longer cycle is not always better. It lowers DoS but raises delay for everyone waiting at red. That is why the lab's optimiser also compares PI and why every region has a maximum cycle.

5 Congestion, exit blocking and gating

IN PLAIN WORDS

A cloakroom with a sensor at the door works fine until the queue for coats reaches the door and sits on the sensor. Then the staff must notice the queue itself. If the cloakroom is about to jam completely, an usher closes the outer door for a moment so the people inside can move.

- Queue over detector: count no longer reliable
- Congestion importance raises a link's weight
- Exit blocking: no room beyond the junction
- Gating stores queues where harmless

0.85 → 1.02

weighted DoS with 20% congestion, importance 1

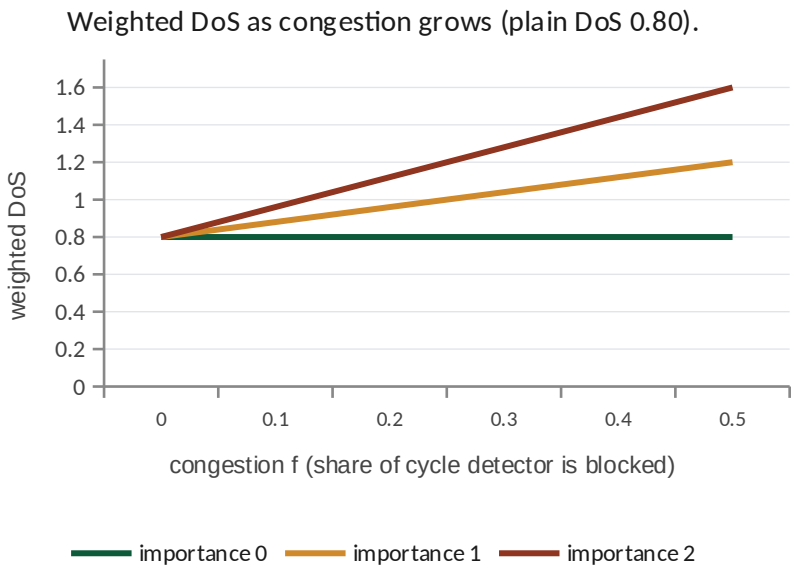
Worked example: A congested link asks for green

congestion f = seconds detector blocked by queue ÷ cycle
weighted DoS = DoS × (1 + importance × f)

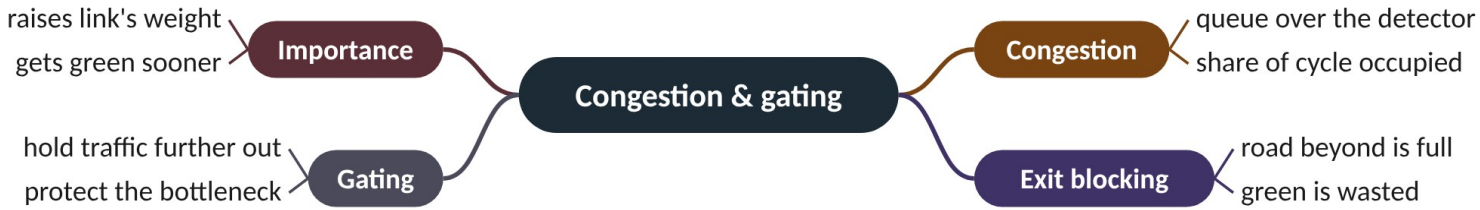
A link has DoS 0.85. In a 60 s cycle its detector is covered by a standing queue for 12 s. Congestion importance = 1.0.

1. Congestion f = 12 ÷ 60 = **0.20**.
2. Weight = 1 + 1.0 × 0.20 = **1.20**.
3. Weighted DoS = 0.85 × 1.20 = **1.02**.
4. Plain DoS 0.85 was under 0.90; weighted 1.02 is **above** it.

The split optimiser now treats this link as the worst one and moves green towards it, before its queue spills back into the junction upstream.



Remember it: Congestion & gating



PICTURE THIS · STOP 5: CLOAKROOM

In the foyer a mountain of coats has piled up over the doorway's motion sensor, which flashes 'BUSY' forever. An usher in white gloves swings the outer gate shut, and outside on the long lawn concert-goers queue happily with picnic blankets.

TRAP

Gating deliberately makes some drivers wait longer. It is worth it only if the queue it creates sits somewhere harmless (a long link) and the bottleneck it protects would otherwise lock up and lose capacity for everyone.

6 Bus priority: extensions and recalls

IN PLAIN WORDS

A host holding a lift door for someone running down the corridor is an extension. Pressing 'close' early to fetch someone waiting on another floor is a recall. Either way, the lift then gets back to its normal routine.

- Buses detected by GPS or detector
- Extension: hold the green a little
- Recall: bring the bus stage early
- Recover the plan in 1-2 cycles

22.9 s

mean bus delay with priority (34.7 s without)

Worked example: Extension or recall?

extension possible if arrival ≤ green end + maximum extension
bus saving = wait without priority − wait with priority

Cycle 80 s. Stage A (bus road) green 0-40 s, intergreen 5 s, stage B green 45-75 s, intergreen 5 s. Max extension 12 s. Minimum green for B 7 s.

1. Bus 1 reaches the stop line at 46 s. Without priority it waits until A returns at 80 s: **34 s**.

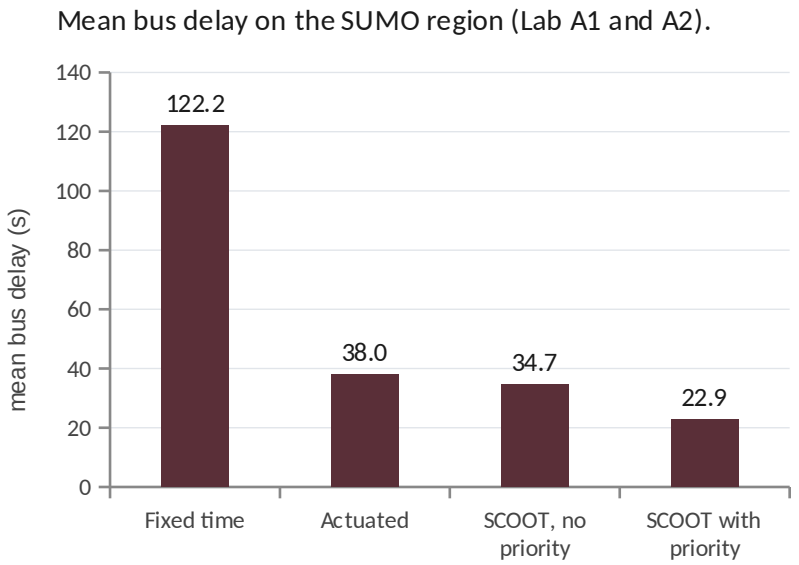
2. 46 ≤ 40 + 12 = 52, so extend A to 46 s: wait **0 s**. Saving 34 s.

3. Bus 2 is detected at 56 s and will arrive at 65 s. Without priority: 80 − 65 = **15 s**.

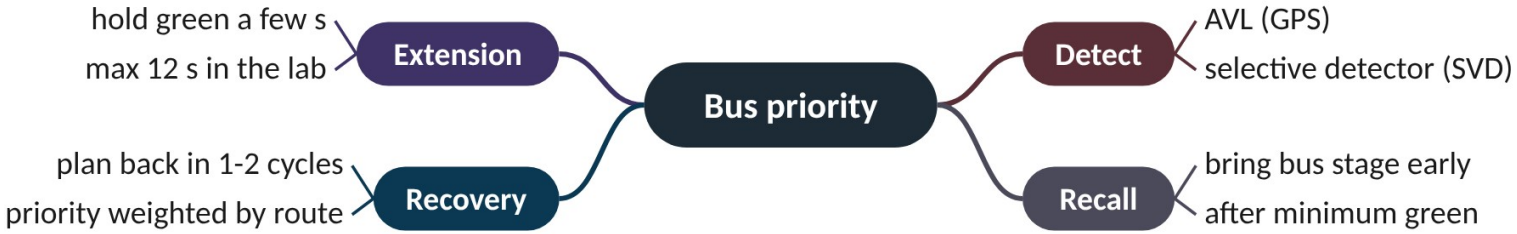
4. Recall: B has had 11 s (≥ 7 s minimum), so cut it at 56 s; after the 5 s intergreen A is green at 61 s.

5. Bus 2 arrives at 65 s on green: wait **0 s**. Saving 15 s.

Extensions help buses that just miss the green; recalls help buses that arrive mid-red. Stage B lost 6 s and 19 s respectively, which SCOOT recovers over the next cycles.



Remember it: Bus priority



PICTURE THIS · STOP 6: BUS IN THE PIT

A red double-decker bus rolls into the orchestra pit. The conductor holds the last violin note on and on until the bus has squeezed past (extension), then later cuts a song short with a crash of cymbals to let a second bus in (recall).

TRAP

Priority is not free. Every second given to the bus stage is taken from another stage this cycle. That is why extensions are capped, recalls respect minimum greens, and recovery restores the plan quickly.

7 The wider SCOOT system

IN PLAIN WORDS

A good doctor keeps your records. When you come in with a temperature, she compares it with your normal readings, not with someone else's. ASTRID is the records cabinet; INGRID is the doctor who says 'this is not normal for you'.

- ASTRID: stores history and profiles
- INGRID: spots incidents from detector patterns
- Emissions, pedestrians, cooperative vehicles
- Same principles live on in newer systems

2020

SCOOT 7: cooperative data and cloud-first design

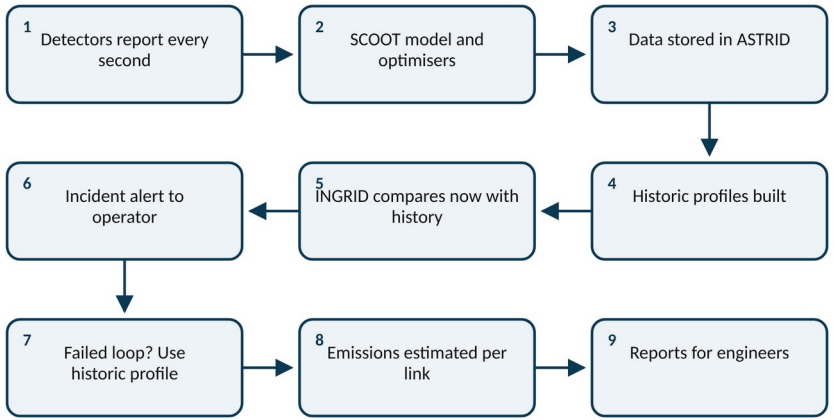
Worked example: Does today look normal?

ratio = flow now ÷ historic flow for this time

At 08:15 on a weekday, ASTRID's historic profile for a link says 60 LPU per 5 minutes. Today the detector shows 12 LPU, and the detector is occupied 85% of the time (normally 20%).

1. Flow ratio = 12 ÷ 60 = **0.20**: one fifth of normal flow.
2. Occupancy ratio = 85 ÷ 20 = **4.25** times normal.
3. Low flow plus high occupancy means traffic is standing, not absent.
4. This pattern is typical of a blockage downstream: **raise an incident alert**.

Low flow alone could mean an empty road at night. Low flow with very high occupancy means a queue that is not moving. Comparing with history is what makes the difference visible.



Remember it: Wider system



PICTURE THIS · STOP 7: ARCHIVE ROOM

Behind the stage is a dusty archive where an archivist named Astrid files a recording of every concert. Her twin sister Ingrid listens to tonight's music through a brass trumpet and rings a fire bell the moment it sounds unlike the recordings.

TRAP

Historic substitution is a fallback, not a fix. A link running on yesterday's profile cannot react to today's surprise, so failed detectors must still be repaired quickly. (Mastery tests exactly this.)

8 The full system on a SUMO region

IN PLAIN WORDS

To learn what each part of a car does, a mechanic removes one part at a time and test-drives it. Remove the steering and you crash; remove the radio and nothing changes. An ablation study does the same with SCOOT's mechanisms.

- Four junctions, tidal peaks, buses every 4 min
- SCOOT: −65% delay against fixed time
- Split optimiser matters most
- Honest surprises: fixed cycle, congestion

−65%

delay vs fixed time on the SUMO region

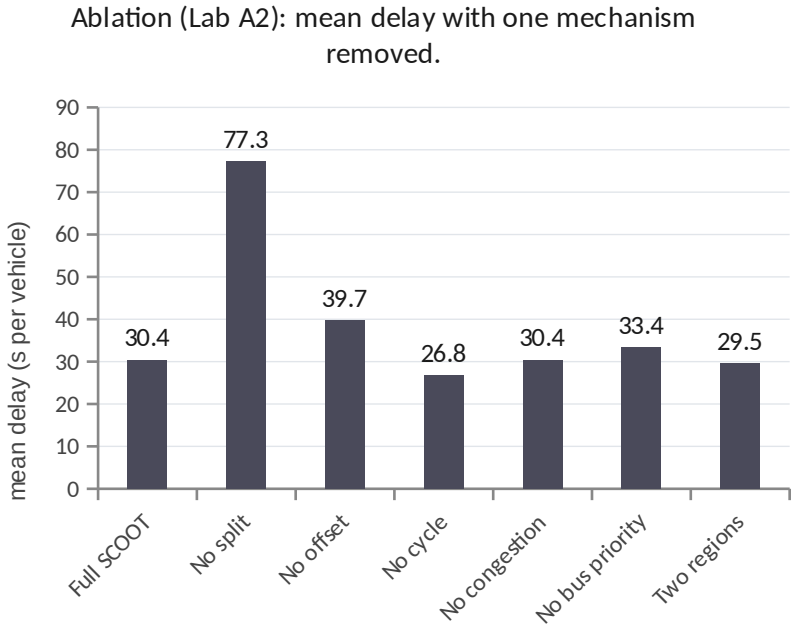
Worked example: Reading the results

change = (new − old) ÷ old × 100%

Lab A1 and A2 mean delays (s per vehicle): fixed 86.95, actuated 29.07, SCOOT 30.44; no split optimiser 77.30. Bus delay: actuated 37.99, SCOOT 22.92.

1. SCOOT vs fixed: (30.44 − 86.95) ÷ 86.95 = **−65%** delay.
2. SCOOT vs actuated: (30.44 − 29.07) ÷ 29.07 = **+4.7%** delay.
3. Bus delay vs actuated: (22.92 − 37.99) ÷ 37.99 = **−40%**.
4. No split optimiser: 77.30 ÷ 30.44 = **2.5 times** the delay.

SCOOT is far better than a fixed plan, about level with good actuation on general delay, and clearly better for buses and stops. The split optimiser is the mechanism that matters most here.



Remember it: Full system



PICTURE THIS · STOP 8: APPLAUSE METER

At the final bow a giant applause meter shows the night's 'grumble' scores: a clockwork robot scores 87, a jumpy rabbit 29, the conductor 30. When the violinists are sent home the grumble leaps to 77; when the metronome is glued at 60, it drops to 27.

TRAP

Do not conclude 'the cycle optimiser is useless'. On this demand the region's maximum (120 s) and slow decreases let the cycle stay too long after the peak. With a tuned maximum it would help; that is why engineers tune each region.

The three optimisers side by side

	Split	Offset	Cycle
Runs	before every stage change	once per cycle per node	every 5 min (2.5 rising)
Changes	one stage change −4/0/+4 s	whole node −4/0/+4 s	region cycle ±4/8/16 s
Chooses by	lowest highest DoS	lowest PI on links in and out	critical node DoS vs 90%
Memory	temporary ±4, permanent ±1	kept until next step	kept until next review
Without it (Lab A2)	77.3 s delay	39.7 s delay	26.8 s (surprise)

Things beginners mix up

This…	…is not this	The fact that tells them apart
Split optimiser	**Offset optimiser**	Split moves one stage change and scores by DoS; offset moves the whole node and scores by PI.
Temporary change	**Permanent change**	Temporary ±4 s applies to this cycle only; permanent ±1 s is stored in the plan.
Extension	**Recall**	Extension holds the bus's green on; recall brings the bus's stage in early.
ASTRID	**INGRID**	ASTRID stores history; INGRID compares now with history to spot incidents.
Double cycling	**A separate region**	A double-cycled node runs exactly 2 cycles per region cycle, so it stays coordinated.
Congestion	**Exit blocking**	Congestion: queue over this link's detector. Exit blocking: no room on the road beyond.

Check yourself (1 of 2)

Greens A 30 s, B 20 s; demand A 20 LPU, B 18 LPU; SATO 1.0. Which split option wins?

A. −4 s

B. 0

C. +4 s

D. No option is allowed

Cycle 60 s, L = 12 s, Y = 0.75. What does the cycle optimiser do?

A. Shorten by 4 s

B. Hold

C. Lengthen by 4 s

D. Lengthen by 8 s

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

In Lab A2, which removed mechanism raised delay most?

A. Offset optimiser

B. Cycle optimiser

C. Split optimiser

D. Bus priority

Flow is 25% of the historic profile and occupancy is 4 times normal. INGRID would most likely:

A. Ignore it: low flow is normal at night

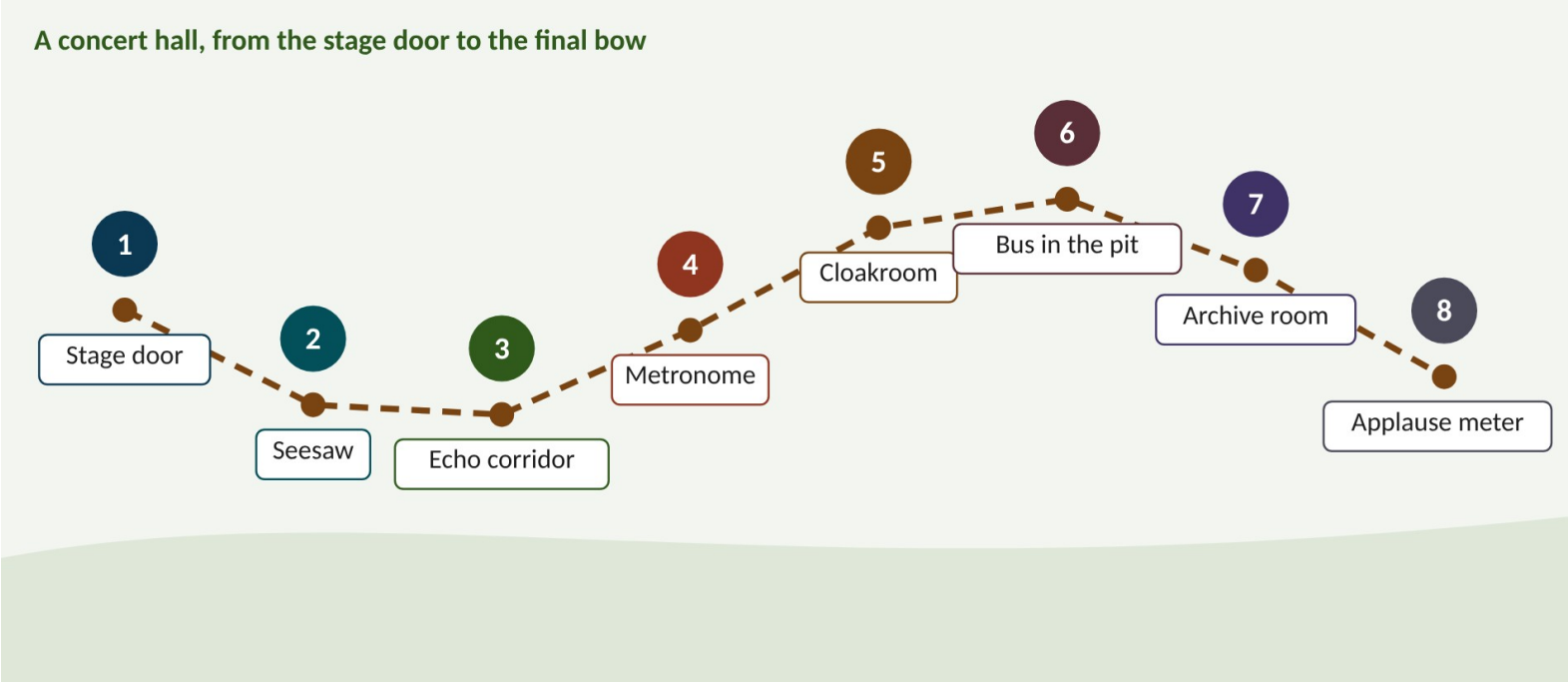
B. Flag a possible incident: traffic is standing

C. Lengthen the cycle

D. Switch the detector off

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Stage door = Regions · 2 Seesaw = Split optimiser · 3 Echo corridor = Offset optimiser · 4 Metronome = Cycle optimiser · 5 Cloakroom = Congestion & gating · 6 Bus in the pit = Bus priority · 7 Archive room = Wider system · 8 Applause meter = Full system

Numbers to remember

› Split: −4/0/+4 s temporary, ±1 s permanent; 88% 'keep' in the lab

› Offset: once per cycle, ±4 s, PI = delay + 4 × stops

› Cycle: every 5 min (2.5 rising), > 0.90 up, < 0.80 down, steps 4/8/16 s

› Double cycling: 96 s → 48 s cuts ped wait 40.3 → 16.7 s

› Bus: max extension 12 s; bus delay 34.7 → 22.9 s

› Lab A1 delay: fixed 86.95, actuated 29.07, SCOOT 30.44 s

› Lab A2: no split 77.3, no offset 39.7, no cycle 26.8 s

Keep going

Study guide: SCOOT_Advanced_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scoot-learning-lab